

SUCCESSIVE DIFFERENTIATION

- Let $f : A \rightarrow R$ where $A \subset R$ be a real valued function. If f is differentiable at every point in A then $f' : A \rightarrow R$ is called the derivative of f or 1st derivative of f .
- If f' is differentiable at every point in A then $f'' : A \rightarrow R$ is called the derivative of f' or 2nd derivative of f .
- Similarly, if 3rd, 4th, ..., (n-1)th derivatives of f exists in A , then $f^{(n)} : A \rightarrow R$ is called as nth derivative of f .
- $f^{(n)}(x)$ = Derivative of $f^{(n-1)}(x)$ where 'n' is a natural number.
- For $n \in N$ and $n > 1$, $\frac{d^n y}{dx^n} = \frac{d}{dx} \left(\frac{d^{n-1} y}{dx^{n-1}} \right)$.

- If a function is given in the parametric form $x = f(t)$ and $y = g(t)$ then $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ and

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}.$$

- If a function is given in implicit form $f(x, y) = c$, then $\frac{dy}{dx} = \frac{-f_x}{f_y}$ where f_x and f_y are the partial derivative of f with respect to x and y .
- $D^n((ax+b)^m) = m(m-1)(m-2)\dots(m-n+1).a^n(ax+b)^{m-n}$ where m is a rational number and n is a positive integer.
- $D^n((ax+b)^m) = \frac{m!}{(m-n)!}.a^n(ax+b)^{m-n}$ where m, n are positive integers and $m > n$.
- $D^n((ax+b)^m) = n!.a^n$ where m, n are positive integers and $m = n$.
- $D^n((ax+b)^m) = 0$ where m, n are positive integers and $m < n$.
- $D^n\left(\frac{1}{ax+b}\right) = \frac{(-1)^n n!.a^n}{(ax+b)^{n+1}}.$
- $D^n\left(\frac{1}{(ax+b)^2}\right) = \frac{(-1)^n (n+1)!.a^n}{(ax+b)^{n+2}}.$
- $D^n\left(\frac{1}{(ax+b)^r}\right) = \frac{(-1)^n (n+r-1)!.a^n}{(r-1)!. (ax+b)^{n+r}}.$
- $D^n(\log|ax+b|) = \frac{(-1)^n (n-1)!.a^n}{(ax+b)^n}.$
- $D^n(e^{ax+b}) = a^n.e^{ax+b}.$

- $D^n(a^{bx+c}) = b^n \cdot (\log a)^n \cdot a^{bx+c}$.
- $D^n(\sin(ax+b)) = a^n \cdot \sin\left(\frac{n\pi}{2} + ax+b\right)$.
- $D^n(\cos(ax+b)) = a^n \cdot \cos\left(\frac{n\pi}{2} + ax+b\right)$.
- $D^n(e^{ax} \sin(bx+c)) = \left(\sqrt{a^2+b^2}\right)^n \cdot e^{ax} \sin\left(bx+c+n+\tan^{-1}\frac{b}{a}\right)$.
- $D^n(e^{ax} \cos(bx+c)) = \left(\sqrt{a^2+b^2}\right)^n \cdot e^{ax} \cos\left(bx+c+n+\tan^{-1}\frac{b}{a}\right)$.
- $D^{2n}(\sin(ax+b)) = (-a^2)^n \cdot \sin(ax+b)$.
- $D^{2n}(\cos(ax+b)) = (-a^2)^n \cdot \cos(ax+b)$.
- $D^n\left(\frac{ax+b}{cx+d}\right) = \frac{(bc-ad)(-1)^n n! c^{n-1}}{(cx+d)^{n+1}}$.
- $D^n(x^n \log x) = n! \left(\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}\right)$.
- $D^n(x^m \log x) = \frac{(-1)^{n-m-1} (n-m-1)! m!}{x^{n-m}}$. Where $n > m$.
- $D^n\left(\frac{\log x}{x}\right) = \frac{(-1)^n n!}{x^{n+1}} \left(\log x - 1 - \frac{1}{2} - \frac{1}{3} - \dots - \frac{1}{n}\right)$
- $D^n(f(x) \pm g(x)) = D^n(f(x)) \pm D^n(g(x))$.
- $D^n(kf(x)) = kD^n(f(x))$, where k is any constant.
- **Leibnitz's Theorem:** If u, v are two functions of n^{th} derivatives such that uv possesses n^{th} derivative then $D^n(u.v) = {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + {}^nC_n u v_n$.